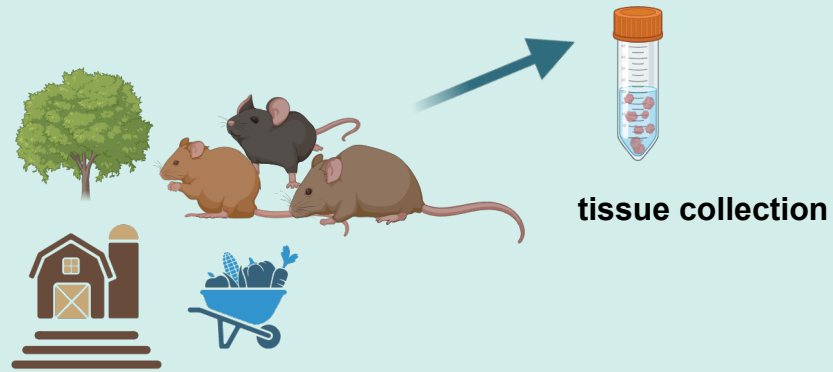


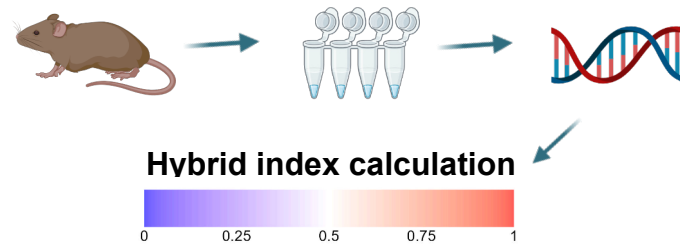
a) Field sampling:

335 wild mice from 120 localities (2016-2019)

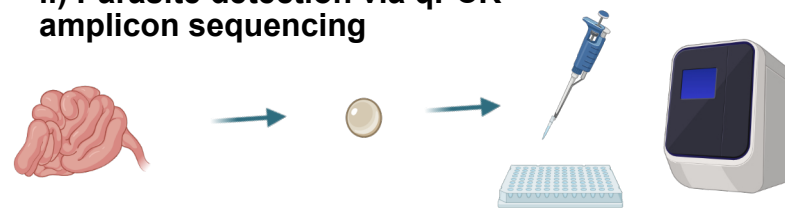


b) Molecular characterisation:

(i) Host genotyping with 14 diagnostic markers

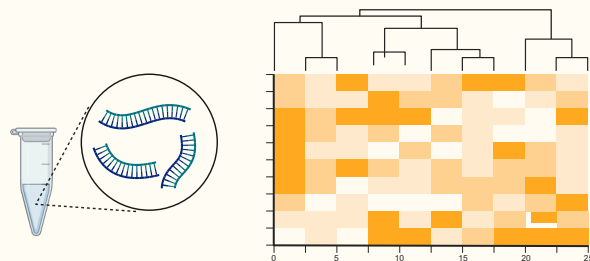


(ii) Parasite detection via qPCR + amplicon sequencing



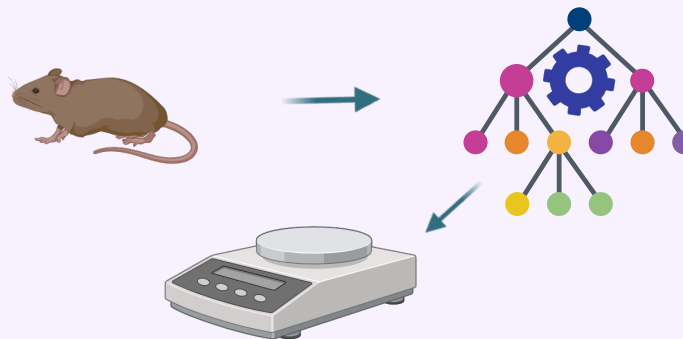
c) Immune profiling:

- RNA extraction
- cDNA synthesis
- Fluidigm qPCR of 19 immune genes
- quantile normalisation
- MICE imputation.



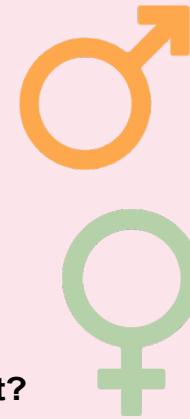
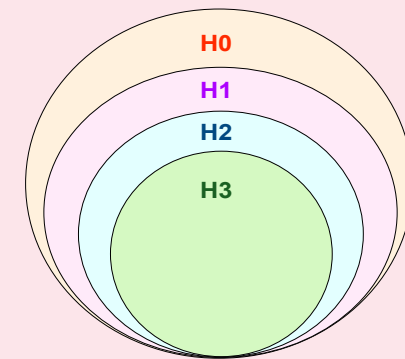
d) Predictive modelling:

Apply Random Forest model
trained on 136 laboratory mice,
47% variance explained



predicted weight loss (3.98-18.5%).

e) Statistical validation: Sex-specific hybrid effects



Key Comparisons:

- H1 vs H0: Subspecies effect?
- H2 vs H0: Group effect?
- H3 vs H2: Interaction effect?